LAB NO.:	**1:Snort Installation**	**SEM/TRADE:**	SEM 4 / CBS 1
INDIVIDUAL/ GROUP MEMBERS:	**1. ABDUL HARIZ BIN MOHD SARNUDIN**	**TTO:**	**Mohamad Aiman Hanif**
		MARKS:	**/20**

German-Malaysian Institute **CIT 0592 |Intrusion Detection & Prevention Systems**

(5%)

		MARKS				
No.	**Criteria**	**NEEDS IMPROVEMENT [1]**	**APPROACHES EXPECTATIONS[2]**	**MEETS EXPECTATIONS [3]**	**EXCELLENT [4]**	
GROUP/INDIVIDUAL LABWORK						**Marks**
1.	**Understa n din g on Topology of the scenario**	Cannot develop understanding on the SNORT requirement	Able develop understanding on the SNORT requirements but still not accurate and detail	Clear, accurate understandi n g explanation of the SNORT requirement with diagrams and complete but with NO example	Clear, accurate understanding explanation of the SNORT requirements and complete example.	

2	**Knowledge on lab objectives and concepts**	Students failed to demonstrate clear understanding of the lab objectives and concepts.	Students demonstrates some knowledge of the lab objectives and concepts.	Student demonstrates sufficient level of the lab objectives and concepts.	Student demonstrates well verse of the lab objectives and concepts.	
3	**Lab completion**	Not able to achieve any of lab objectives and show any output.	Able to achieve some lab objectives and show some output.	Able to achieve most of lab objectives and show most of the output.	Able to completely achieve all lab objectives and show all of the output.	
4	**Interpreting Findings**	Students failed to interpret the findings and result.	Student able to interpret some of the findings and result, but in a somewhat unclear manner.	Students able to interpret most of the findings and result.	Student able to interpret all the findings and result clearly and comprehensively.	
5	**Level of Needed assistance**	Student was unable to complete task with assistance.	Student was unable to complete task without major assistance.	Student was able to complete the task with little assistance.	Student was able to complete the task without assistance.	

German-Malaysian Institute | All rights reserved. No part of this document may be utilized in any form without permission from the Institute.

TOTAL MARKS:	/20	

ASSESSOR COMMENT:	

Answer all questions:

1. Explain the basic requirement of the SNORT installation. You may use a diagram/screenshot to support your explanation.

   Linux Operating System Platform: A pristine Linux environment—such as the Ubuntu deployment utilized in this lab—to host the installation framework and execute deployment commands.

   Packet Capture Core (libpcap): A foundational library that enables Snort to intercept, ingest, and extract raw data packets directly from the physical or virtual network interface.

   Data Acquisition (DAQ) Module: A specialized abstraction layer that interfaces with the packet capture library, ensuring Snort can process line-rate network traffic efficiently.

   Parsing & Decompression Utilities: Essential dependency libraries, including PCRE for analyzing regex-based rule patterns and ZLIB for decoding and inspecting compressed data payloads.

   Promiscuous Mode Configuration: The requirement for the network interface card (NIC) to operate in promiscuous mode, allowing the IDS to sniff all transit traffic across the network segment rather than just packets destined for this specific virtual machine.

   Local Subnet Definition: The exact identification of network boundaries using CIDR notation (e.g., `192.168.145.0/24`), which is necessary to define the internal home network within Snort's configuration files.

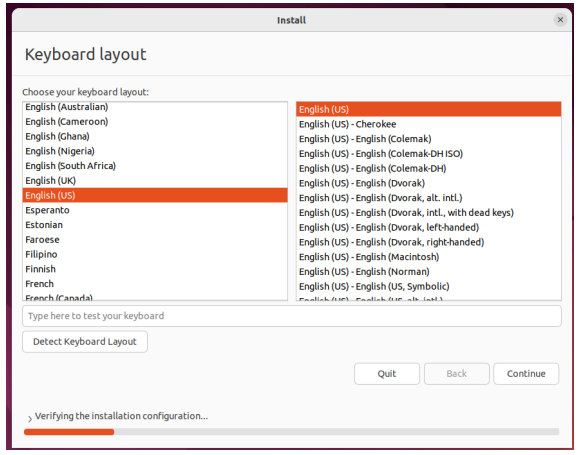
2. Explain your expectation from this lab. What is your achievement before and after lab completion.

Phase	Before Lab Completion	After Lab Completion
OS & Linux Command Line	Possessed a foundational awareness of Ubuntu but lacked hands-on experience managing and configuring network utilities via the CLI.	Developed strong proficiency in updating system repositories and seamlessly executing administrative and deployment commands.
Snort Engine Setup	Recognized Snort conceptually as an industry-standard IDS, but lacked the technical knowledge to compile it or define local network boundaries.	Successfully deployed Snort (v2.9.15.1) alongside all prerequisite dependencies and core background libraries.
Network Interface Tuning	Understood standard IP addressing principles but was unfamiliar with the mechanics of configuring a virtual network interface card (vNIC) for packet capture.	Identified the active interface (ens33), mapped it to the local subnet parameters, and successfully enabled promiscuous mode for full packet sniffing.

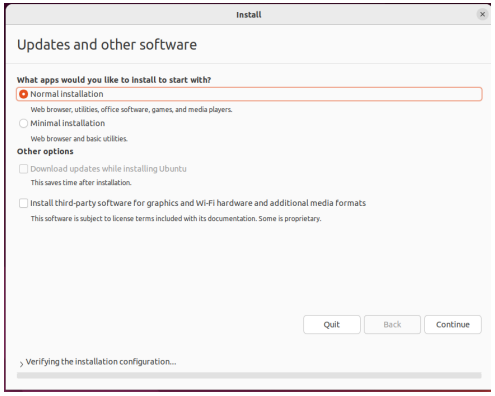
3. Did you able to complete the lab successfully (YES/NO)? Explain why and how. Demonstrate and explain the output you achieve from this lab.(Explanation at least in 10 point

The Snort deployment was completed successfully without encountering any critical configuration errors. The operational workflow involved updating core system packages, executing the standard package manager deployment, defining network parameters during the post-install prompt, and validating the active state of the service.

SNORT INSTALLATION

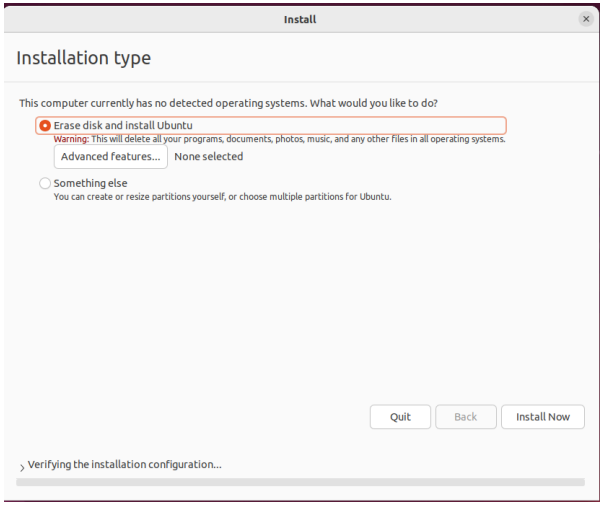


Choosing a Normal Installation guarantees that native system tools, networking libraries, and utilities are bundled alongside the operating system shell



Running snort --version confirms that the engine is fully compiled and dynamically linked with libpcap 1.10.1 and PCRE 8.39. This validates that the deployment is fundamentally capable of intercepting raw link-layer network packets and parsing intricate, regex-based threat signatures without throwing runtime syntax errors.

Executing 'Erase disk and install Ubuntu' cleanly prepares the primary hard drive system structures, defining the default paths



The execution of `sudo apt update` demonstrates an active internet connection

```
beel@beel-virtual-machine:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for beel:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
```

`apt install snort` output indicates that the engine successfully

```
beel@beel-virtual-machine:~$ sudo apt install snort -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libdaq2 libdumbnet1 libluajit-5.1-2 libluajit-5.1-common libnetfilter-queue1
  oinkmaster snort-common snort-common-libraries snort-rules-default
```

verified that the machine handles an active operational Ethernet

```
beel@beel-virtual-machine:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.145.131  netmask 255.255.255.0  broadcast 192.168.145.255
        inet6 fe80::10fe:b83b:c3d5:c995  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:88:f2:18  txqueuelen 1000  (Ethernet)
        RX packets 486620  bytes 722869566 (722.8 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 30988  bytes 2301772 (2.3 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

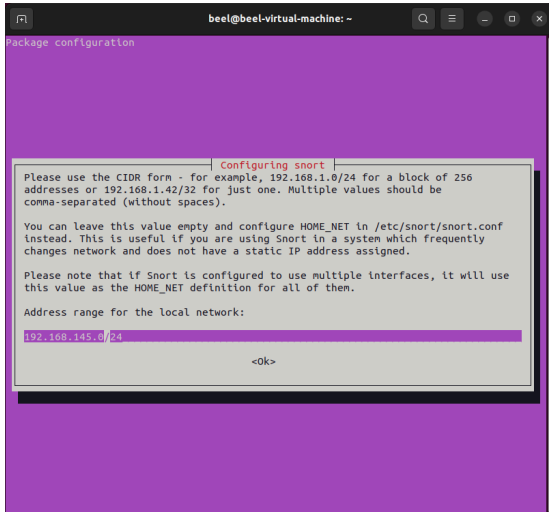
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 420  bytes 47276 (47.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 420  bytes 47276 (47.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Defining the local network boundary as 192.168.145.0/24 allows Snort to accurately differentiate between internal lateral traffic (East-West) and external incoming traffic (North-South). Properly scoping this variable is a critical prerequisite for baseline accuracy, as it prevents a high volume of false positives by distinguishing routine internal communications from external scanning attempts.

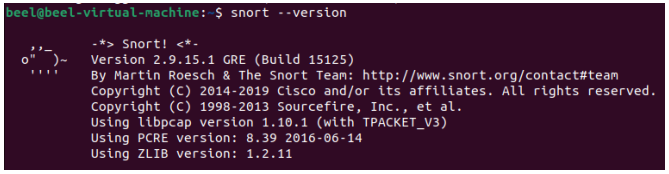
The network adapter profile explicitly shows a configured IPv4 address of **192.168.145.131**

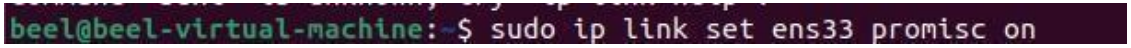
The terminal printout details a netmask configuration of **255.255.255.0**

The package configuration interface records the successful binding of **192.168.145.0/24** directly into the system



Running `snort --version` successfully prints out the official ASCII art banner, confirming that Snort version



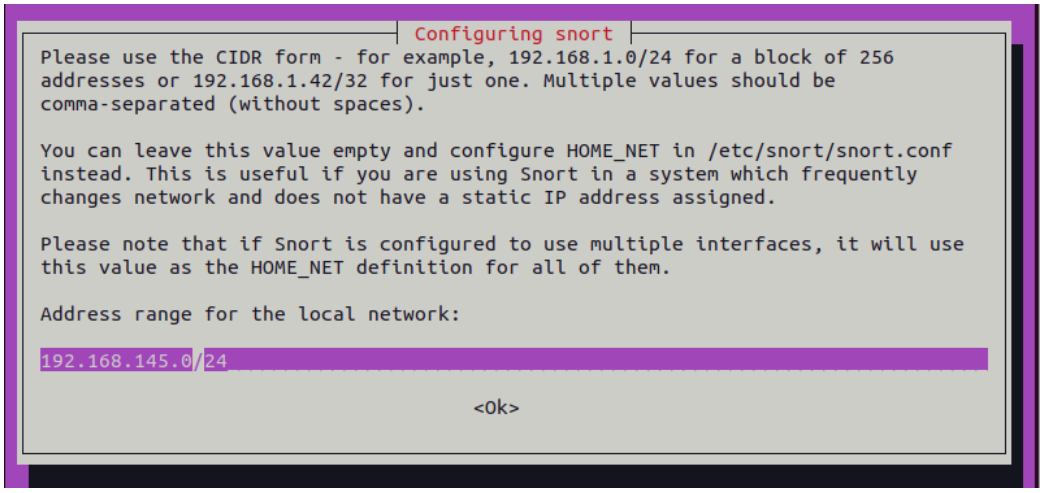


4. Interpret the result and your findings? You explanation may include the result of your SNORT and how you elaborate the finding

Executing the snort --version command confirms that the application binary is properly compiled and dynamically linked with its core dependencies: libpcap 1.10.1 and PCRE 8.39. This diagnostic output serves as proof that the installation is functionally sound, possessing the underlying hooks required to intercept raw link-layer traffic and process complex regular expression threat signatures without encountering runtime errors.



Configuring the local network boundary to 192.168.145.0/24 provides the engine with necessary contextual awareness. This parameter allows Snort to explicitly differentiate between benign internal communications (East-West traffic) and external probing or inbound attack vectors (North-South traffic). Properly defining this home network variable is crucial for baseline data hygiene, as it drastically reduces the volume of false-positive alert.



Enforcing promiscuous mode using the sudo ip link set ens33 promisc on command is what elevates the deployment from a passive host listener to a fully functional network monitor.

Standard network interface cards (NICs) operate on a strict filtering mechanism, automatically discarding any network frames not explicitly addressed to their unique MAC address. Activating promiscuous mode overrides this hardware-level filter, instructing the interface to capture and forward all transit traffic within the broadcast domain directly to Snort for deep packet inspection.